Fiscal Year:	FY 2014	**Task Last Updated:**	FY 03/10/2014
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	216-433-2859
Organization Name:	NASA Glenn Research Center		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2014, per PI (Ed., 10/17/13) NOTE: Extended to 12/31/2013, per PI (Ed., 10/10/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirsch, David (NTEC, Jacobs Engineering)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA STD 6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 psia. Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro-, or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth. It will improve the safety of astronauts in spaceflight so they can return safely to Earth.
Task Progress:	In FY 13, we successfully passed the Science Concept Review and completed a Science Requirements Document for the flight experiment. These are significant milestones on the path to flight hardware development. I also successfully completed a number of 5.2 second drop tower tests with PMMA rods and cheesecloth fabric studying flame spread and flammability limits. In the drop tower, I was able to quantify the blowoff branch of the extinction curve, but the time is too short to obtain quenching extinction. In the cheesecloth tests, I tested a variety of area densities to determine the flame spread rate dependence on thickness for concurrent flame spread, to compare it to opposed flow inverse dependence. I am very interested in determining limits for both opposed and concurrent flame spread to determine which configuration is the worst case in spacecraft fires. As part of this project, I defined 20 samples to be flown for BASS-II in the ISS MSG. We built and launched these samples, and testing has begun. On the second test, I was able to observe a quenching extinction process, which is quite different from the blowoff extinction. In blowoff extinction, the flame tip opens up and then the flame blows downstream to extinguish. In contrast, for quenching extinction, the flame shrinks toward the stagnation region (at the tip of the rod) and gets smaller and smaller until it disappears. I was selected to participate as a Co-Investigator on a JAXA 2012 AO Selected experiment studying Fire Safety. I have been involved in defining the science requirements for the JAXA experiment hardware to be tested in the Kibo module on ISS. Lastly, I defined a Normoxic Upward Pressure Limit (NULP) for NASA Test 1 tests, which takes advantage of the normoxic atmospheres for exploration missions to simplify the parameter space, in terms of pressure and oxygen, to evaluate the flammability limits of materials in Test 1. I also included a factor of safety on this NULP to account for the enhanced flammability in space.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Olson S. "Radiative Exchange during Concurrent Flame Spread in Microgravity." 2014 Spring Technical Meeting of the Central States Section of The Combustion Institute, Tulsa, OK, March 16-18, 2014. 2014 Spring Technical Meeting of the Central States Section of The Combustion Institute, Tulsa, OK, March 16-18, 2014. http://www.cssci.org/meetings/2014tulsa/program.php , Mar-2014
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Abstracts for Journals and Proceedings	Kikuchi M, Mizushima T, Fujita O, Nakamura Y, Takahashi S, Ito A, Torikai H, Olson SL. "Overview of the Solid Combustion Experiment in the Japanese Experiment Module." 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 104. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Ferkul PV, Olson SL, Johnston MC, T'ien JS. "Flammability Aspects of Fabric in Opposed and Concurrent Air Flow in Microgravity." 8th U.S. National Combustion Meeting, University of Utah, Salt Lake City, UT, May 19-22, 2013. 8th U.S. National Combustion Meeting, University of Utah, Salt Lake City, UT, May 19-22, 2013. , May-2013
Abstracts for Journals and Proceedings	Olson SL, Lee JR, Fujita O, Kikuchi M, Kashiwagi T. "Quantitative Infrared Image Analysis of Simultaneous Upstream and Downstream Microgravity Flame Spread over Thermally-Thin Cellulose in Low Speed Forced Flow." 8th U.S. National Combustion Meeting, University of Utah, Salt Lake City, UT, May 19-22, 2013. 8th U.S. National Combustion Meeting, University of Utah, Salt Lake City, UT, May 19-22, 2013. , May-2013
Papers from Meeting Proceedings	Olson SL, Clayman L, Kacher H, KuczmarskiMA, Gokoblu SA. "Pressure Response in Enclosures During and After Large-Scale Flame Spread: Testing and Modeling." 43rd International Conference on Environmental Systems, Vail, CO, July 14-18, 2013. 43rd International Conference on Environmental Systems, Vail, CO, July 14-18, 2013. http://dx.doi.org/10.2514/6.2013-3412 , Jul-2013